Received: 12 November 2025 | Revised: 2 June 2026 | Accepted: 8 June 2026
DOI: 10.1002/ppp3.70259

RESEARCH ARTICLE

Plants People Planet PPP Open Access

Soil health assessment using buried cotton underpants with the help of 1000 citizen scientists

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Funding information
Ernst Göhner Stiftung, Grant/Award Number: 2020-1112/3; Stiftung Dreiklang; Stiftung Mercator Schweiz, Grant/Award Number: 2020-4120; Participatory science academy UZH/ETH, Grant/Award Number: PWA 02/2020; Paul Schiller Stiftung, Grant/Award Number: 4707; Mercator foundation; Swiss National Science Foundation, Grant/Award Number: 194126; Paul Schiller Foundation; Foundation Dreiklang; Participatory Science Academy

Societal Impact Statement

Soil is a vital resource for life on Earth. Although many soils are degraded, societal awareness of soil-related issues is limited. To highlight soil as a key resource and fascinating study system, we initiated a citizen science project together with 1000 participants investigating soil health across Switzerland using buried cotton underpants. The degree of underpants decomposition was interpreted as a measure for soil biological activity and soil health. This work demonstrates that underwear decomposition varied widely across land-use types and that buried underwear is a simple and easily interpretable tool to gain a first impression of soil health.

Summary

- Soil is a vital resource for plant growth. It hosts a major part of global species diversity and is key for ecosystem functioning and it forms the basis of human civilization. Yet soils are rapidly degrading, threatening food production, biodiversity, and Earth system functioning. This threat is largely unrecognized in broader society, hindering effective steps towards soil protection.
- In a citizen science approach, involving 1000 citizen scientists burying more than 2000 cotton underpants and 12,000 teabags across 1000 locations, we aimed to assess soil biological activity and soil health in Switzerland. Decomposition of underpants and tea bags was considered as proxies for soil biological activity and soil health.
- Underwear decomposition differed greatly across land-use types, indicating significant variation in soil biological activity among sites. Land use was the most important predictor of decomposition processes, followed by soil chemical parameters. Private gardens, the land-use type with the highest levels of soil organic carbon

and nutrients, exhibited the greatest biological activity, whereas lawns showed the lowest, with croplands and grasslands falling in between.

- Overall, this project demonstrates that underpants can serve as a simple and easy-to-interpret tool, providing informative insights into the state of soils. Moreover, they are perfectly suited to raise attention in society for soil. We propose the "Underpants Index" as an accessible, engaging method to assess soil decomposition processes and soil fertility and raise public awareness for soil as a vital, yet threatened resource.

KEYWORDS

awareness, citizen science, land-use, soil health, soil your undies, Tea Bag Index, Underpants Index

1 | INTRODUCTION

Soil forms the thin layer that covers large parts of terrestrial surfaces on Earth, supporting the growth of higher plants and, therefore, providing the basis of terrestrial life. It also forms the basis for agricultural production with approximately 95% of the food we eat originating from soil (Montanarella et al., 2015). In addition, soil stores more carbon (C) than is bound in living biomass and in the atmosphere combined (Lal, 2008) and harbors more than half of the biodiversity worldwide (Anthony et al., 2023) (Table 1a). Despite its fundamental importance for life on earth, soil remains largely unnoticed by the general public (Brevik et al., 2019). Soil is often considered as "dirt" (Montgomery, 2012), and many are unaware of the important and fascinating habitat it represents. Farmers, responsible for the management of vast areas of land, often receive little education in basic soil principles (Pauli et al., 2016). The lack of relationship between the public and soil leads to wasteful treatment of global soil resources (Table 1b). Increased awareness and targeted action towards the protection of soils are urgently needed.

Citizen science is an approach involving laypeople in data collection, which allows to expand research activities and to make science more accessible (Fraisl et al., 2022; Fritz et al., 2019). It has the advantage that, through the help of citizens, a much higher number of samples and a wider range of sites can be investigated than through research staff alone. Moreover, citizen science approaches can greatly serve to raise awareness for specific topics. The need for more awareness raising for soils through citizen science has been voiced (Pino et al., 2022), and promoting citizen science approaches for soil monitoring is among the main objectives of the Mission «A Soil Deal for Europe» which is a 1 billion research program from the European Union (Panagos et al., 2024).

Soil health is defined here as the ability of soil to function as a living system, supporting the productivity, diversity and environmental services of terrestrial ecosystems. Soils are considered healthy when they are able to perform key ecosystem functions at high levels, which requires an active soil biological community. Moreover, soil properties, such as soil organic carbon (SOC), cation exchange capacity, and

TABLE 1 Important soil functions (a), soil threats (b), and their global relevance.

a.		
Soil functions	**Relevance**	**References**
Food provision	95% of our food depends on soil	FAO (2013); Montanarella et al. (2015)
Biodiversity	Soils harbor ~59% of global biodiversity	Anthony et al. (2023)
Carbon storage	2nd largest carbon reservoir on Earth (~2,700 Pg); Soil stores more carbon than the atmosphere + living biomass combined; acts as carbon sink/source.	Jobbágy and Jackson (2000); Lal (2008); Amundson et al. (2015); Crowther et al. (2019)
Nutrient cycling	Soil organisms decompose organic matter, recycle nutrients, and drive biological nitrogen fixation. Key for plant growth.	Hopkins and Dungait (2010); Crowther et al. (2019); Anthony et al. (2023)
Water storage	Key to the global water cycle; Soil often stores more water than rivers and lakes combined.	Güntner et al. (2007); Hirmas et al. (2018)
Filtering and cleaning	Soils filter, buffer, and degrade contaminants, protecting ground- and drinking water.	Keesstra et al. (2012); Lamichhane et al. (2016)
Human health	Source of antibiotics (e.g., penicillin); soil helps suppress pathogens.	Wall et al. (2015); Brevik et al. (2020)
b.		
Soil threats	**Relevance**	**Reference**
Soil erosion	20–50 Gt soil/year lost globally; up to 10–20 t/	Montanarella et al. (2015); Borrelli et al. (2017)

(Continues)

TABLE 1 (Continued)

b. Soil threats	Relevance	Reference
	ha/year in ecosystems with poor plant cover.	
Carbon loss	Conversion to cropland causes 16–50% soil C loss; humans have caused ~133 Gt soil C loss as CO_2/CH_4, contributing to climate change	Scharlemann et al. (2014); Sanderman et al. (2017); Beillouin et al. (2023)
Soil sealing	250–300 km^2/day sealed globally under asphalt/ concrete (~40,000 soccer fields/day).	Blum (2013); Gardi et al. (2015); Sarneel et al. (2024)
Soil pollution	Widespread pollution: sources include industry, mining, fossil fuels, pesticides, fertilizers, antibiotics, microplastics (e.g., tire wear), PFAS, wastewater. Soil pollution is poorly quantified; linked to >500,000 premature deaths/year.	Landrigan et al. (2018); Rodríguez-Eugenio et al. (2018); Silva et al. (2019); Khan et al. (2021); Larsson and Flach (2022); Ackerman Grunfeld et al. (2024);(Köninger et al., 2026)

balanced pH values, are closely linked to definitions of soil health (Lehmann et al., 2020; Raghavendra et al., 2020; Romero et al., 2024). A key ecological soil process performed by a wealth of soil organisms is decomposition, through which complex organic matter, such as leaves, wood, or cadavers, are broken down into simpler inorganic and organic compounds and contributing significantly to the formation of soil organic matter, while releasing CO_2 and nutrients (Chapin et al., 2002). Decomposition thus plays a central role in global carbon and nutrient cycles and supports continued ecosystem productivity. Human land-use strongly affects soil biological communities (Labouyrie et al., 2023; Tsiafouli et al., 2015) and has also been shown to affect decomposition processes (Fu et al., 2022; Liu et al., 2020). Although studies comparing decomposition across different land-use systems exist, they are typically limited to few sites or land-use types (Englmeier et al., 2023; Fu et al., 2022; Liu et al., 2020). Large-scale decomposition experiments across different land-use types, for example, at the national scale, are rare, largely due to logistical constraints. For instance, a study investigating global decomposition patterns was based on only 30 sites (Wall et al., 2008). As soil decomposition is driven by soil biological communities (Nannipieri & Badalucco, 2024), it can serve as a proxy for soil biological activity.

Decomposition can be assessed using a wide range of organic materials. Cotton strips have previously been applied (Harrison et al., 1988; Nachimuthu et al., 2007), while the Tea Bag Index (TBI) makes use of commercially available teabags with standardized composition for the same purpose (Keuskamp et al., 2013). Due to the wide availability of the material, the TBI has been employed in a citizen science approach to gather global decomposition data (Sarneel et al., 2024). However, despite being relatively easy to apply and interpret, the method requires careful handling and specific tools, such as a fine scale balance. Moreover, it lacks visual appeal and may struggle to attract broader public attention for soil-related topics. A more playful approach that has occasionally been used to illustrate soil decomposition, though without scientific rigor, is the burial of cotton underpants (Figure 1). Beyond providing a sensory experience of decomposition and soil biological activity, this approach has potential to engage society and gather quantitative data through citizen science (Bender & van der Heijden, 2025; Takáts et al., 2026).

We invited 1000 citizen scientists to quantify underpants decomposition in soil at the national level across major land-use types. We provided standardized underpants and teabags to all participants together with detailed instructions on how to bury them (Figure S1). Participants collected soil samples for chemical and physical analyses and provided site information and management data. All samples, underpants and teabags were centrally analyzed by the research team.

We hypothesized that

- The extent of underpants decomposition reflects soil biological activity and is higher in soils showing chemical characteristics (i.e., increased SOC, balanced pH, etc.) that are indicative of higher soil health.
- Tea bag decomposition, as assessed through the TBI, is significantly correlated to the level of underpants decomposition.

The project was launched under the title "Proof by underpants," and the extent of decomposition was termed the "Underpants Index" (UI). The project received global media attention on radio, television, in newspapers, and the internet (e.g., media in over 25 countries reported on this project) and could clearly communicate the importance of soil and the need for improved soil protection measures (Bender & van der Heijden, 2025). Moreover, it provided ample opportunities to bring soil related topics into art (Underpants Dress, Audio S1). Here, we demonstrate that underpants decomposition can serve as a simple and easy-to-interpret tool to assess decomposition and provide a first impression of soil health. We observed that underpants decomposition differed among land use types and was highest in private gardens, a land use type that often shows high soil fertility.

2 | MATERIALS AND METHODS

2.1 | Citizen science approach

Before project launch, a call for participation was disseminated through Swiss media and collaborating organizations. Around 1400 people registered within a few days. Because participation was limited

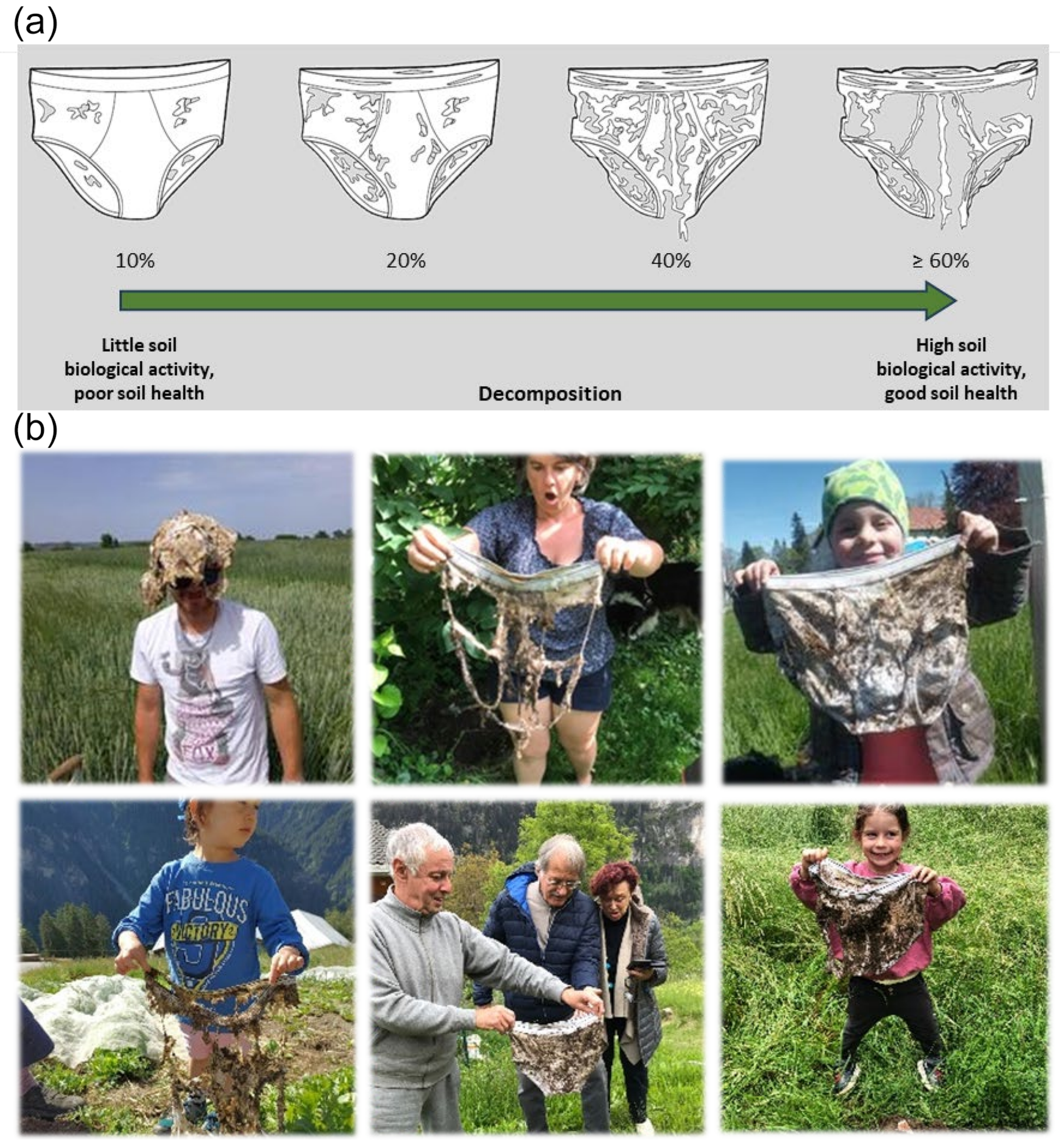

FIGURE 1 (a) Basic hypothesis as communicated to citizen scientists in the project. The more the underpants are decomposed after 1, resp. 2 months, the more active the soil biological community. (b) Participants at the moment of retrieving underpants from soil. All people shown in the pictures, or their legal representatives, have given written consent for the pictures to be published. Photographs by: Manuel Steinemann, Cécile Dauny, Rémy Jobin, Christian Hunger, Luisa Poggi, and Yann Hautier.

to approximately 1000 individuals, participants were selected to ensure a geographically balanced distribution across Switzerland (Figure 2), with at least 50% being farmers.

A smartphone app and website were developed as a central information and data collection hub, allowing participants to enter site and management information and upload geo-referenced photos of sites and underpants to an online map (Beweisstück Unterhose-SPOTTERON Citizen Science).

After registration, participants received an activity pack by mail containing two pairs of organic cotton underpants, 12 teabags, a sampling bag for soil collection, and detailed burial and sampling instructions (https://soilproof.org/anleitung_beweisstueckunterhose_mit_unterhosenset/; Figure S1). Instructions were provided in German, French, and Italian, the three main Swiss national languages. Packs were shipped with a prepaid return box to facilitate returns.

2.2 | Materials

2.2.1 | Underpants

A total of 2000 organic cotton underpants corresponding to a product sold by the Swiss supermarket chain Coop (Coop Naturaline, men's briefs, white, in size S, art. no.: 3305289013) were ordered directly

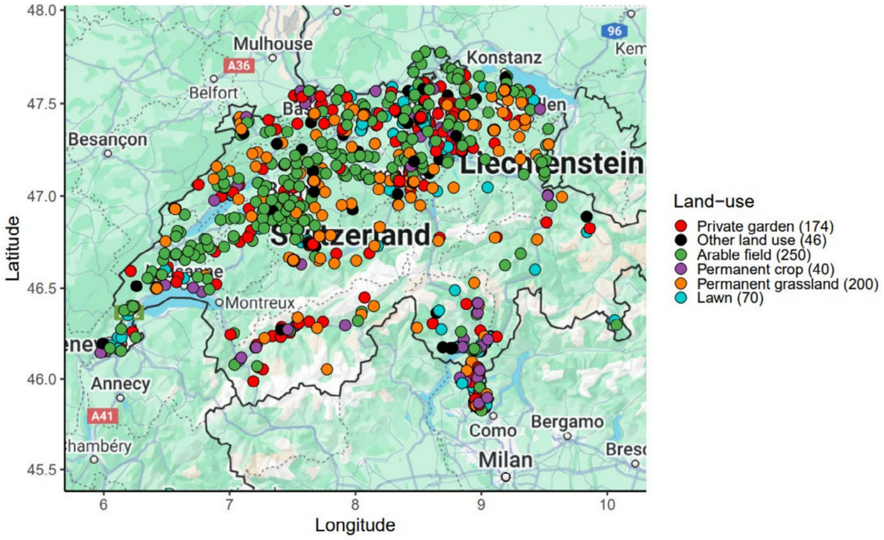

FIGURE 2 Map of Switzerland showing the location and land-use of sites where data for this study were collected. After data cleaning and filtering, 780 locations remained for analysis. Underwear was mainly buried in areas where population density is higher and rarely in the mountains (e.g., the central part of Switzerland). Numbers in brackets indicate the number of sites for each land-use type.

from the producer. Underpants consisted of 100% organic cotton except for waistband and seams, which consisted of synthetic materials. This approach ensured the use of a standardized organic material throughout the experiment.

2.2.2 | Teabags

To calculate the TBI, two types of teabags were used as standardized plant litter following the TBI protocol: A rooibos tea ("Rooibos & Hibiscus infusion", EAN8722700188438, Lipton Tea Unilever, Dublin, IE) and a green tea ("Sencha Exclusive Selection", EAN 8722700188438 Lipton Tea Unilever, Dublin, IE) both in non-woven polypropylene bags. These specific tea blends have been chosen because of the specific composition of the bag, made from synthetic material resisting decomposition, their specific mesh size, and their potentially global commercial availability (Keuskamp et al., 2013).

2.3 | Burying of underpants and teabags and collection of soil samples

Participants selected a site of their choice, dug a hole of approximately 60 × 30 × 30 cm (Figure S1bA), and collected a soil sample (Figure S1bB). Two underpants were inserted vertically next to each other so that the waistband remained visible above the soil surface. The hole was then partially refilled until both leg openings were covered (corresponding to a depth of 8 cm below the soil surface), and three teabags of each tea type were placed in front of each underpants (Figure S1bC). The remaining soil was replaced, compacted by trampling, and the location marked with a yellow wooden marker (Figure S1bD).

After 30 days, the first pair of underpants and associated teabags was retrieved (Figure S1bE). After 60 days, the second underpants and remaining teabags were collected. All materials were air-dried and mailed to Agroscope for analysis (Figure S1bF).

2.4 | Management data

A smartphone app was developed in collaboration with Spotteron (www.spotteron.net, Vienna, AT) to allow participants to register exact site locations, provide site background and management information, and upload photos of the site and decomposed materials to an online map (https://soilproof.org/map). The app was available in German, French, Italian, and English. For participants unable or unwilling to provide data digitally, a paper questionnaire containing the same core questions was provided. Questionnaires differed slightly between farmers and hobby gardeners, with farmers being asked for more detailed agronomic information.

2.5 | Sample processing

Upon arrival at the research station, underpants and teabags were dried at 70°C for 24 h and soil samples at 40°C for 24 h to ensure complete drying. Soil samples were then sieved to 2 mm prior to laboratory analysis.

2.6 | Soil analyses

Chemical soil properties were analyzed by the analytical chemistry laboratory at Agroscope according to Swiss reference methods (Eidgenössische Forschungsanstalten FAL, 1996) as described in detail in the Supporting Information (Methods S1). Ranges of measured soil parameters are shown in Table S1. In some cases, insufficient soil material was available for all analyses, and selected parameters were not measured, resulting in different sample sizes across variables (Methods S1).

2.7 | Underpants processing and index calculation

Underpants decomposition was quantified both as weight loss and area loss. For each pair of underpants, debris and adhering soil were removed by hand; however, some soil particles remained, which may have reduced the accuracy of weight-based estimates.

For area-based assessment, underpants were cut open along the sides, spread on a blue background, and photographed vertically from a fixed height. Images were analyzed in ImageJ (Schneider et al., 2012). The blue background was removed using an RGB filter, and the remaining underpants material was quantified in pixels. Area loss was calculated relative to the area of new underpants and expressed as percentage decomposition:

$$\begin{aligned}\text{Underpants Index (UI)} &= \text{decomposition}\,(\%)\\ &= (A_{new} - A_{decomp})/A_{new}{}^{*}\,100,\end{aligned} \tag{1}$$

where A_{new} is the area of new, undecomposed underpants, and A_{decomp} is the area of respective underpants being buried for 1 or 2 months in the soil.

Mass loss and area loss were strongly correlated ($R^2 = 0.73$, $p < 0.001$; Figure S2), indicating that both metrics can estimate decomposition. However, area-based decomposition was used for subsequent analyses because it was considered more reliable.

2.8 | Tea bag processing and TBI calculation

Teabags were cleaned of stones and adhering soil and weighed using a precision scale (0.001 g). For each site and sampling event, average values were calculated for rooibos and green tea using all available replicate bags (up to three per tea type). Damaged teabags were excluded.

The TBI method uses two litter types with contrasting decomposability to estimate decomposition dynamics from a single time-point measurement. According to the original protocol (Keuskamp et al., 2013), TBI is assessed after 90 days. In this study, retrieval occurred after approximately 30 and 60 days. Analyses therefore focused on TBI calculated after 60 days as this more closely approximates the published protocol.

TBI parameters were derived from tea weight loss following Keuskamp et al. (2013). The TBI consists of two indices: the litter stabilization factor *S* and the decomposition rate constant *k*.

$$TBI\,(S) = 1 - (a_g/H_g), \tag{2}$$

where a_g is the fraction of decomposed green tea and H_g is the hydrolyzable fraction of green tea (= 0.842; see Keuskamp et al. (2013)).

Assuming equal litter stabilization for both tea types, the decomposable fraction of rooibos tea (a_r) was calculated as follows:

$$a_r = H_r\,(1 - S). \tag{3}$$

where H_r is the hydrolyzable fraction of rooibos tea (= 0.552, see Keuskamp et al. (2013)).

The decomposition rate constant of rooibos tea was then calculated as follows:

$$TBI\,(k) = ln\,(a_r/(W_{(t)} - (1 - a_r)))/t \tag{4}$$

where *t* is time in days and *W*(*t*) is the remaining rooibos tea mass after decomposition and where *t* is the time in days and $W_{(t)}$ the remaining weight of rooibos tea after decomposition.

Since TBI (k) represents a decomposition rate and therefore carries similar information to the percentage underpants decomposition over time, this parameter was used for furthering our analysis.

2.9 | Categorizations of Management Data

Collected site and management variables included land-use type, current vegetation, winter vegetation cover, type and amount of fertilization, type and maximum depth of soil treatment, weed and pest control, use of solid organic fertilizers during the previous 5 years, and irrigation. Fertilizer categories are described in Methods S1.

The data indicated that plant protection strategy and soil tillage were strongly nested within land-use type, with private gardens and permanent grasslands generally lacking tillage or pesticide application, while arable systems typically included both. In addition, many participants left some questions unanswered, substantially reducing sample size for certain variables. We therefore focused mainly on land-use type and fertilizer type, for which sample sizes remained sufficient.

Samples were excluded if the key variable land-use type was missing or if more than 75% of variables were NA. An overview of land-use categories is provided in Table S2.

2.10 | Climate data and elevation

Daily temperature and rainfall data for May–July 2021 were obtained from all Swiss weather stations via MeteoSwiss (Swiss Federal Office of Meteorology and Climatology). Each site was assigned data from the nearest weather station. Elevation was extracted from geographical coordinates using the digital elevation model DHM25 (Federal Office of Topography swisstopo, Wabern, CH). An overview of elevation and climatic variables is provided in Table S3.

2.11 | Statistics

Of the 1100 participation packs sent out, 883 soil samples, 1752 underpants, and 7088 intact teabags were returned, together with 800 questionnaires containing background and management data. After quality filtering, 780 samples remained for statistical analysis. Because not all variables were available for all samples, the dataset still contained missing values, and a conditional listwise deletion strategy (entire observation excluded if variables in statistical model contain one or more NA values) was used, resulting in different sample sizes for different analyses. Statistical analyses were performed with R version 4.3.0 (R Core Team, 2023).

To identify the most important variables influencing underpants and tea bag decomposition, we first performed multiple regression analyses. Strongly correlated explanatory variables (correlation coefficient >0.8) were removed beforehand. The best performing regression models were selected from all possible combinations of explanatory variables using the package glmulti (Calcagno & de Mazancourt, 2010), and predictor importance was assessed using relaimpo (Groemping & Matthias, 2018).

Because missing data substantially reduced the number of observations available for the final multivariable models, we additionally tested numerical predictors individually using non-parametric Spearman rank correlations and assessed categorical variables (land-use and fertilizer type) using mixed-effects models. In these models, elevation was included as a fixed covariate and soil texture class as a random effect using lme4 (Bates et al., 2011). These variables were included because they were identified as influential but are not directly affected by management. Post hoc analyses were performed using the emmeans package (Lenth & Lenth, 2018).

UI and TBI values were logit-transformed to meet model assumptions, which were checked graphically using the package performance (Lüdecke et al., 2021). A principal component analysis (PCA) was conducted on centered and standardized soil variables. Site scores were visualized in ordination space and colored either by land-use category or the underpants decomposition grade, with variable loadings overlaid. Differences in multivariate soil composition among land-use categories were tested using PERMANOVA based on Euclidean distances. Spearman rank correlations between ordination axes and underpants and tea bag decomposition were also performed.

Figures were created using ggplot2 (Wickham et al., 2023), and maps were generated using RgoogleMaps (Loecher & Ropkins, 2015) and ggmap (Kahle et al., 2019).

3 | RESULTS

3.1 | Factors affecting underpants and tea bag decomposition

A total of 1000 citizen scientists buried 2000 underpants and 12,000 teabags and collected soil samples across Switzerland (Figure 2).

The UI (i.e., the percentage of underpants decomposition) was generally lower after 1 month (10.1% ± 0.56%) than after 2 months (50.45% ± 0.86%, Figure S3a,b). Multiple regression analyses identified land use, soil nutrient concentrations (available P [P_{avail}] and sodium [Na^+]), fertilizer type, and soil texture (silt content) as the strongest predictors. Variables related to water availability (average rainfall and irrigation), soil pH, and latitude were also recognized as relevant, although not significant (Table S4a and Figure 3a). Overall,

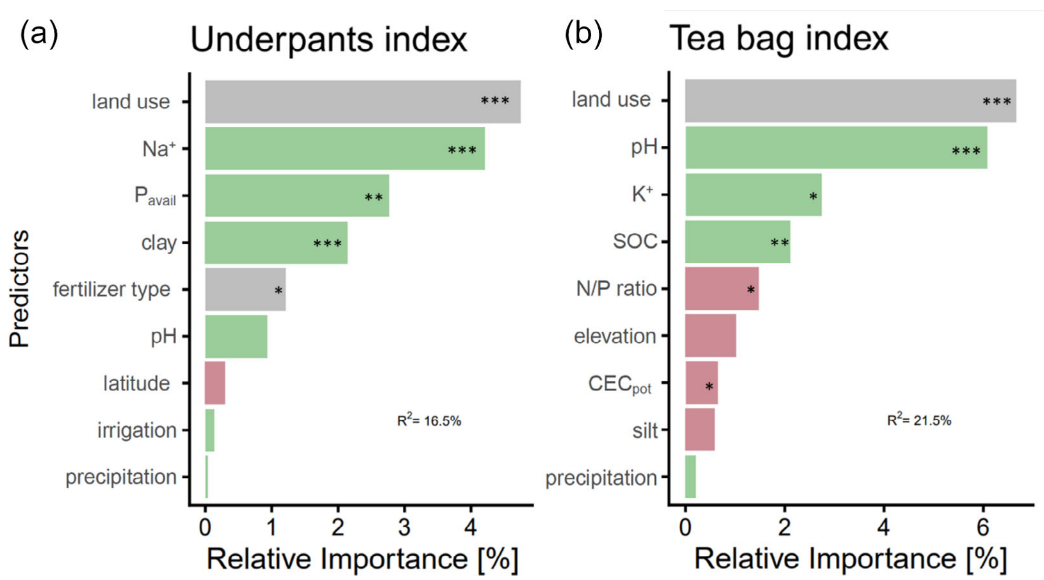

FIGURE 3 Relative importance (share of model R^2 explained) of variables explaining (a) the Underpants Index and (b) the Tea Bag Index as assessed through multiple regression models. Green bars indicate a positive relationship between predictor and decomposition; red color indicates a negative relationship. Gray color indicates categorical variables; the statistical significance of the different variables is represented as *p < 0.05; **p < 0.01; ***p < 0.001. See Table S4 for details.

the assessed variables could explain 16.5% of the total variation in the UI after 2 months. Fertilizer types were not homogeneously distributed across land-use types with private gardens often receiving compost and permanent cultures and lawns often receiving fertilizers from the category “other”. No fertilizer was applied in sites categorized as “Other land-use” (Figure S4).

The TBI was higher after 1 month than after 2 months. This was expected, since the easily degradable fraction of the tea material decomposes more rapidly (Figure S3e,f). Land use, pH, soil organic carbon (SOC), and a range of soil chemical parameters (available Potassium [K_{avail}], N/P ratio, potential cation exchange capacity (CEC_{pot})] were identified as significant explanatory variables for the TBI. Moreover, elevation, precipitation, and silt content were included in the model, although not showing significant effects (Table S4b, Figure 3b). Overall, the model explained 21.5% of variation in the TBI. The UI and the TBI were overall significantly correlated (spearman's rho = 0.32, $p < 0.001$, Figure S5), indicating that both methods indicate similar processes.

3.2 | Effect of land use

Land use had a strong effect on the UI ($F_{5,\ 696} = 6.77$, $p < 0.001$, Figure 4a). On average, private gardens exhibited the highest decomposition after 2 months with around 58% decomposed. In contrast, lawns showed the lowest decomposition, averaging around 40%. Both differed significantly from arable fields that showed intermediate decomposition levels of around 51% and from permanent grasslands with decomposition levels of around 47%. Lawns also showed significantly lower decomposition than the “other land-use” category. The remaining land use types did not differ significantly from each other (Figure 4a).

Land use also had a strong effect on the TBI ($F_{5,\ 518} = 13.87$, $p < 0.001$, Figure 4b), though the observed pattern differed slightly from the underpants data. Private gardens showed the highest decomposition, and arable fields showed significantly lower average TBI values compared to private gardens. Permanent crops showed similar decomposition to arable fields but did not differ significantly from private gardens. Permanent grasslands showed significantly lower decomposition rates than private gardens and arable fields, while lawns showed significantly lower decomposition rates than private gardens but not arable fields.

3.3 | Effect of fertilizers

Fertilizer category had a significant effect on the UI ($F_{4,\ 611.3} = 3.63$, $p = 0.006$, Figure S6a), and decomposition was on average highest in sites where compost was applied, followed closely by mineral fertilization, manure, and slurry. The category “other” showed significantly lower decomposition compared to compost application. For the TBI, sites with compost application showed significantly higher decomposition compared to manure, slurry, and other ($F_{4,\ 472} = 6.6$, $p < 0.001$, Figure S6b).

3.4 | Soil parameters

Land-use types showed significant differences in most soil parameters assessed (Table S5). Private gardens and the “other land use” category had the highest average SOC values at around 3.4% ($F_{5,\ 718} = 16.8$, $p < 0.001$, Figure 4c). In contrast, arable fields and lawns had the lowest SOC values averaging around 2.1%, while permanent crops and grasslands showed intermediate values. P_{avail} levels also varied significantly between land-use types ($F_{5,\ 705} = 44.3$, $p < 0.001$, Figure 4d). Private gardens exhibited average levels of 13.74 mg P_{avail} kg^{-1}, which was significantly greater than in all other land-use types and roughly fourfold higher than what was observed in arable fields. The lowest *P* availability was observed in permanent grassland sites. For a range of further soil parameters, significant differences were also observed (Table S5). PERMANOVA indicated significant differences in multivariate soil composition among land-use categories ($R^2 = 0.09$, $p = 0.001$). Given the high P_{avail} levels and the soil N_{tot}/P_{tot} ratio of 2.38 in private gardens, comparable to arable fields (Table S5), it can be assumed that *P* availability was not limiting plant growth and, hence, additional P fertilization has no effect on plant yield. Correlations of the UI and TBI with the measured soil and environmental parameters are presented in the Supporting Information (Results S1; Tables S6 and S7 and Figure S8).

4 | DISCUSSION

Using a nationwide citizen science approach, this study assessed soil biological activity by measuring the decomposition of buried cotton underpants in a standardized way across various land-use types.

Moreover, it tested whether underpants decomposition can serve as an indicator for soil health.

Nearly 1000 participants buried underpants in a standardized manner. This project shows that the UI can serve as a simple, effective tool offering insights into soil characteristics with implications for soil fertility and health. Findings aligned with the established Teabag Index (TBI). We therefore propose the UI as an easily interpretable tool to investigate soils for scientists and nonspecialists alike. Additionally, underpants decomposed in soil provide a sensory experience of soil activity, making them an excellent outreach tool (Figure 1).

Land-use emerged as the most important factor influencing both UI and TBI, followed by various soil and environmental parameters. However, the multiple regression analyses explained only a relatively small portion of the variation (16.5% for UI and 21.5% for TBI, respectively) suggesting that several key factors may not have been captured in the current analysis. Such low explained variance is common in field experiments and with large and complex ecological datasets, as shown for the human gut microbiome (Johnson, 2020; Jokela et al., 2023) and soil microbial communities (Labouyrie et al., 2023), where substantial variation among sites and individuals is expected under non-standardized conditions. The citizen science approach added further variability through missing values in the data, differences in sampling, adherence to instructions, and questionnaire interpretation. Additional

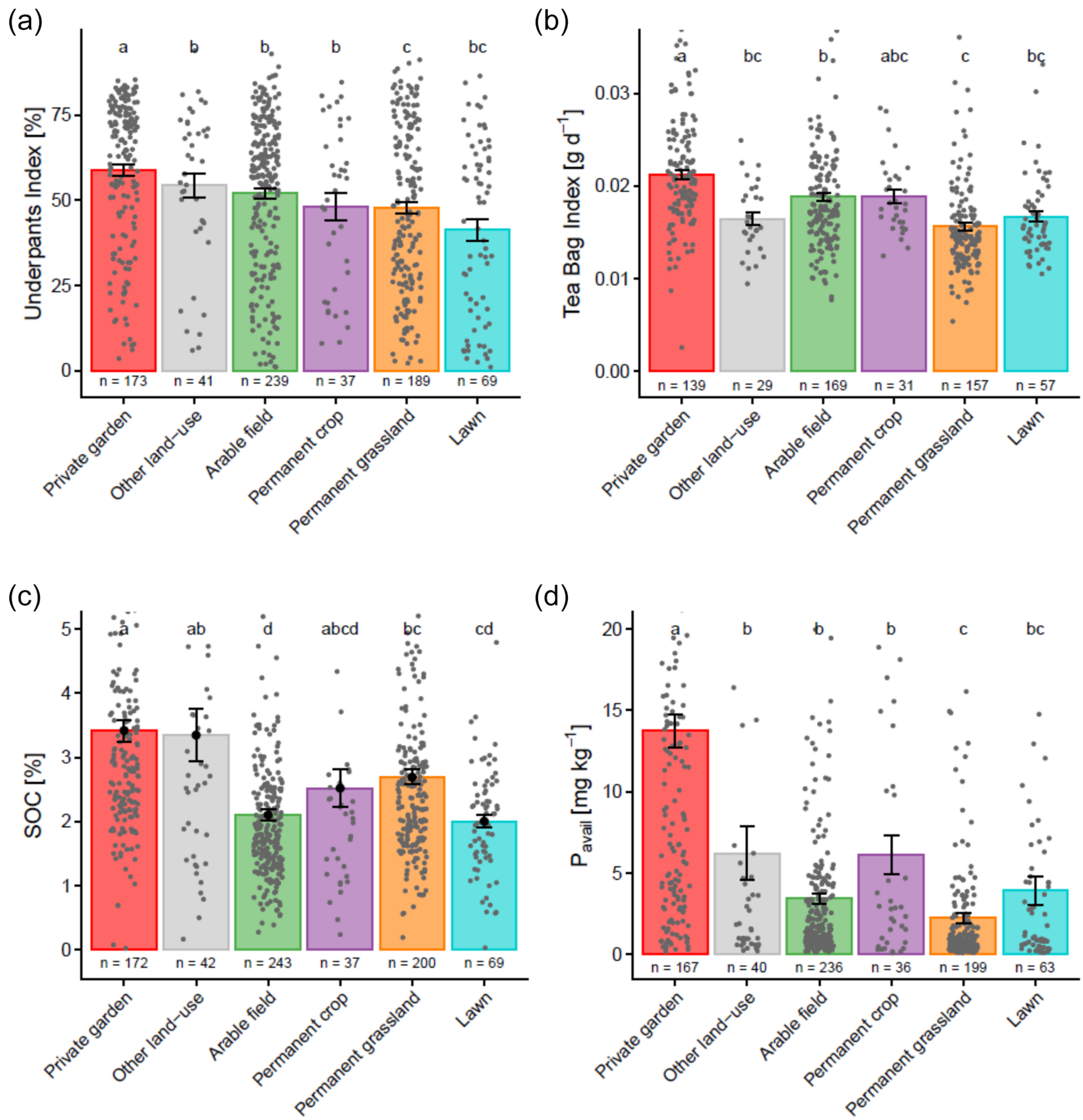

FIGURE 4 Effects of land use type on (a) the Underpants Index and (b) the Tea Bag Index assessed after 2 months and on (c) soil organic carbon (SOC) and (d) available phosphorus concentrations in soil across 780 sites in Switzerland. Numbers below bars indicate the number of samples obtained for each land-use type. Not all analyses could be performed for all sites, for example, if not enough sample was provided or if teabags or underpants could not be retrieved. Error bars represent ±1 SE. Different letters indicate significant differences at $p < 0.05$. In graphs (b–d), outliers were cropped from the graph and the y-axis was truncated for improved representation. Graphs with the full data are shown in Supporting information Figure S7a–c.

soil indicators such as aggregate stability or soil biological community composition across all sites may have provided further insights but were beyond the scope of this project.

We hypothesized that underpants decomposition is highest in soils showing chemical characteristics, such as high soil organic carbon and high nutrient availability, indicating better soil health. The UI was indeed highest in private gardens, which also had the highest SOC and nutrient concentrations, and lowest in lawns, which showed the lowest SOC concentrations. Arable fields showed intermediate decomposition levels, although their SOC concentrations were similarly low to those in lawns. This difference may be related to greater physical soil disturbance in arable systems, which can stimulate microbial activity, decomposition, and SOC loss (Lal, 2004; Zhang & Yu, 2020). Lawns, depending on their type, are often characterized by frequent mowing, low plant diversity, and regular herbicide use, all of which can negatively affect soil biological communities (Cheng et al., 2008; Köninger et al., 2026; Watson et al., 2020). At the same time, lawns remain under continuous plant cover and are usually physically undisturbed, providing relatively stable conditions for soil organisms (Mathew et al., 2012). Private gardens may, like arable fields, experience regular soil management and thus represent another comparatively disturbed land-use system. In addition, they had the highest nutrient concentrations of all land-use types, which likely further promoted decomposition of carbon-rich substrates such as cotton underpants (Bonanomi et al., 2017). A citizen science study conducted in Hungary also found a trend towards higher underpants decomposition in private gardens (Takáts et al., 2026).

Underpants decomposition decreased with elevation, likely due in part to lower temperatures at higher altitudes. In addition, farmland management intensity in Switzerland is generally higher at lower elevations (Klaus et al. 2023), implying that nutrient concentrations are also likely to be greater, potentially promoting decomposition. Most private gardens are also probably located in lower elevations, where population density is higher.

In line with our hypothesis, underpants decomposition increased with SOC contents, soil nutrient concentrations, and soil pH (Table S6), chemical parameters having established connectivity to definitions of soil health (Lehmann et al., 2020; Raghavendra et al., 2020). The UI showed stronger correlations with soil nutrient concentrations and cation exchange capacity than the TBI (Table S6).

In the multiple regression analyses, fertilizer type and P_{avail} concentrations were significant predictors of the UI but not of the TBI. This may be because green and rooibos tea leaves contain higher nutrient concentrations (e.g., K, Na, and other micronutrients) (Malongane et al., 2020; Szymczycha-Madeja et al., 2012), making their decomposition less sensitive to nutrient limitations than cotton underpants, which consist largely of low-nutrient cellulose fibers (Middleton et al., 2021). This suggests that the UI may be a more sensitive indicator of soil fertility than the TBI.

Although SOC is often considered a key indicator of healthy and biologically active soils, especially for managed mineral soils (Lehmann et al., 2020), it was not identified as a significant predictor in the multiple regression model. Because SOC was positively correlated with most measured nutrient parameters, the model suggests that nutrient availability may be more important than SOC itself for underpants decomposition. Taken together, this suggests that soil fertility rather than overall soil health is a driver of underpants decomposition.

According to Swiss fertilization guidelines, optimal concentrations of extractable soil phosphorus (measured in CO2-saturated water) range from 1.6 to 4.3 mg kg^{-1}, depending on soil texture (Flisch et al., 2009). In our data, arable fields averaged 3.5 mg kg^{-1}, while private gardens averaged 13.4 mg kg^{-1} with some values exceeding 60 mg kg^{-1} (Figure S7c). This suggests that many gardeners fertilize well beyond what is necessary for optimal crop growth (Wielemaker et al., 2019). Such excess phosphorus cannot be fully taken up by plants and may be lost through erosion or leaching, thereby negatively affecting groundwater, surface water, and biodiversity (Carpenter et al., 1998; Sharpley et al., 2001). High phosphorus concentrations can also adversely affect soil organisms, particularly beneficial arbuscular mycorrhizal fungi (Treseder, 2004; Treseder & Allen, 2002; van der Heijden et al., 2015). These findings highlight the need for stronger outreach to private gardeners on careful nutrient management. Our results indicate that in most cases, no additional P fertilizer is required to maintain plant growth.

Many private gardens reported compost use, but it remains unclear whether the high phosphorus concentrations were caused by compost alone or by additional nutrient inputs.

Multiple fertilizers were often applied together, making it difficult to separate their effects. Mineral fertilizers frequently combine N, P, and K in one product. Private gardens showed not only the highest P_{avail} but also the highest N_{tot} and K_{avail} values, suggesting that combined NPK fertilizers may be commonly used. However, given the limited management information available, conclusions regarding fertilizer effects should be interpreted with caution.

4.1 | The UI, a soil health indicator?

This project aimed to assess decomposition and soil biological activity across Switzerland using buried cotton underpants and to test whether the UI is comparable to the TBI and suitable as an indicator of soil health. Cotton decomposition is biologically driven, and our analyses suggest that soil nutrient concentrations are important for this process. These findings imply that the UI can provide useful information about soil fertility and associated biological decomposition activity.

Decomposition tended to be highest in land-use systems where soils are regularly managed and likely exposed to greater physical disturbance, such as private gardens and arable fields. Although decomposition was highest in private gardens, which also had significantly higher SOC, nutrient concentrations showed stronger relationships with decomposition than SOC. Combined with the observation that nutrient levels, especially P_{avail}, were highest in private gardens and often far exceeding agronomic thresholds, our analysis suggests that the UI primarily reflects soil fertility.

This has important implications for interpretation. In arable fields and private gardens, higher decomposition and thus potentially greater fertility may be considered beneficial. In contrast, in many natural ecosystems such as meadows or forests, high nutrient availability is often undesirable as it is linked to reduced plant diversity and the disappearance of rare plant species (Clark et al., 2017). To identify the precise drivers of underpants decomposition, controlled experimental studies are needed, for example, by directly manipulating soil fertility, compost application, or SOC levels.

Overall, UI and TBI showed very similar relationships with soil chemical and physical properties, elevation, moisture, and with each other. This suggests that both indicators provide equally meaningful information about soil conditions. However, while the TBI requires careful cleaning and weighing of teabags and subsequent calculations, underpants provide a direct visual and haptic measure of decomposition that is easy to interpret. Although decomposition was quantified here using relatively complex image-based analyses, visual re-evaluation of our data using a degradation scale in 10% increments proved to be fast, simple, and similarly accurate (Figure 5). The UI can

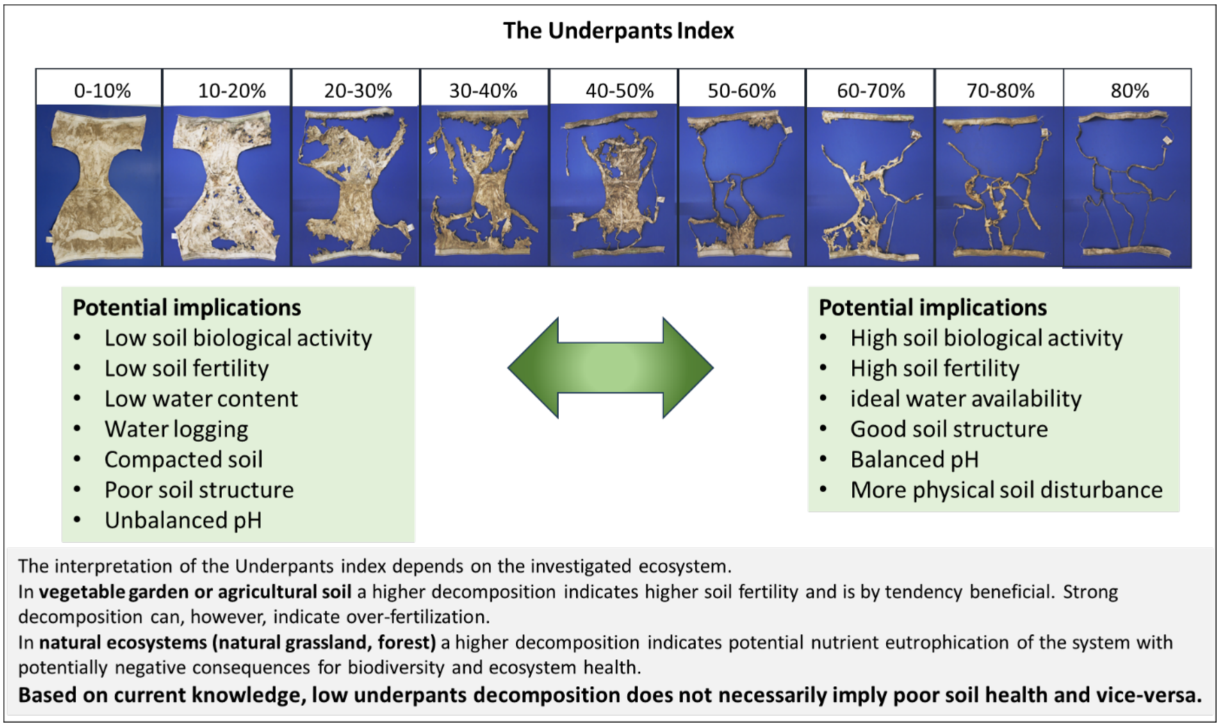

FIGURE 5 The Underpants Index can provide a visual and easily interpretable insight into a soil's state.

therefore serve as a useful and informative tool for both scientists and lay people to assess soil decomposition activity and fertility. It can deliver valuable insights into decomposition processes across different land-use types, provide easily accessible information without requiring specific skills or equipment, and therefore serve as an ideal tool to be used in citizen science projects and to raise awareness for soil-related issues (Bender & van der Heijden, 2025).

5 | CONCLUSION

In collaboration with 1000 citizen scientists, this study demonstrates that the UI provides a simple and informative proxy for soil biological activity across a wide range of land-use systems. The strong correspondence with the TBI supports its usefulness as a low-threshold indicator for assessing soil decomposition processes and soil characteristics with implications for soil health. Our results suggest that the UI reflects soil fertility, particularly nutrient availability, more strongly than soil health in a broader ecological sense. Owing to its simplicity, intuitive interpretation, and strong outreach potential, the UI represents a valuable tool for both scientific investigations and public engagement with soils.

AUTHOR CONTRIBUTIONS

S.F.B., M.G.A.v.d.H., P.V. and A.B. planned and designed the research. S.F.B., D.B., L.B., E.K., N.P., R.D., S.M., A.I., D.M., P.V., A.B., F.L., S.G., Y.H., P.P., P.G., M.G., D.J., A.V., C.W., S.J., N.G., P.A., H.A., M.A., C.A., B.A., F.A., T.A., R.A., A.A., M.An., B.Ar., O.A., B.Au., V.B., B.B., G.B., S. B., C.B., S.Ba., D.Ba., D.Be., M.B., A.Be., S.Be., Y.B., C.Be., B.Be., M.Be., M.Bes., D.Bi., D.Bic., M.Bi., J.B., R.B., J.Bi., B.Bi., A.Bi., U.B., N.B., Y.Bo., M.Bo., L.Bo., T.B., E.B., C.Br., A.Br., N.Br., J.Br., S.Bu., M.Bu., D.Bu., I.B., M.Bur., A.Brk., V.Bu., A.C., S.C., M.C., E.C., B.C., C.C., D.C., C.Ca., E.C.C., R.C., S.Ca., L.E.G.C., E.Ca., L.C., M.Ch., C.Ch., S.Ch., E.Cl., M.Cl., L.Co., M.Co., H.C., S.Co., C.Co., B.Cu., J.L.D., A.D., F.D., D.D., P.D., B.D., A.Dl., N.D., D.De., F.De., M.D., A.De., C.D., M.Do., S.D., Y.D., J.D., M.Du., L.E., B.E., U.E., D.F., P.F., A.F., J.B.F., U.F., F.G.F., N.F., S.F., G.R.F., J.F., F.F., P.Fr., A.Fr., S.Fr., E.F., A.G., P.Ga., A.J.G., C.N.G., E.G., A.Gi., F.G., S.Gi., S.Go., I.G., F.Gr., U.G., G.G., M.Gu., L.G., M.H.G., C.G., G.Gu., D.G., H.G., S.H., U.H., P.H., M.H., M.Hn., G.H., J.H., K.H., A.H., Z.H., T.H., R.H., R.He., M.He., A.He., S.J.H., C.H., S.He., R.Hes., W.H., S.L.H., C.Ho., J.Ho., M.Ho., T.Ho., M.Hu., C.M.H., D.H., B.L.H., W.Hu., E.I., S.I., D.I., R.C.J., P.J., S.D.J., M.J., J.J., A.J., C.J., B.J., D.Ju., W.K., F.O.K., F.K., A.K., L.K., A.Ka., P.K., J.D.K., J.K., E.Ko., G.K., P.Ko., M.K., M.Ku., M.Kn., L.Ku., D.K., R.L., G.L., A.L., J.L., P.L., M.L., S.L., E.L., C.L., P.Lo., C.Lt., L.M., S.Ma., H.M., A.M., L.Ma., E.M., S.Mau., H.Me., S.Me., V.C.M., S.Mey., M.M., P.M., F.M., V.M., A.Mo., R.M., J.M., B.M., A.Ml., L.Ml., T.N., N.N., M.N., S.O., L.O., S.Ot., S.Ou., A.H.P.P., N.Pa., M.P., A.P., N.Pav., C.S.P., B.L.P., A.Pe., P.P.P., S.P., C.P., L.P., M.Po., L.Po., P.A.P., B.P., L.R., S.R., D.R.R., E.R., B.R., H.R., D.R., R.R., K.R., C.R., H. Ro., P.R., J.R., F.R., R.Ro., F.Ro., A.R., N.R., P.Re., N.A.M.Z.F.R., M.R., E.S., A.S., A.Sc., R.S., B.S., S.S., P.S., B.Sc., J.S., K.S., J.Sc., M.S., S.Sc., R. Sc., T.S., J.Sch., S.Sch., A.Sch., C.S., E.Sc., T.Si., C.Si., R.Si., D.S., I.S., R. So., S.R.S., L.S., M.St., J.P.S., P.St., K.St., M.Su., V.T., C.T., C.Se., S.T., V. To., M.T., P.T., D.T., H.A.U., P.F.U., C.V., D.V., A.v.d.H, S.v.M, L.v.M, T. v.W, C.Va., F.V., M.V., C.Vi., R.V., F.Vo., J.V., A.Vo., T.Vu., R.W., B.W., M.W., J.W., K.W., L.W., G.W., T.W., T.Wt., N.Z., A.Z., P.Z., A.Zu., R.C.Z., T.D.B., and M.G.A.v.d.H. performed the research and collected the data. D.B., T.D.B., R.D., N.P., A.I., D.M., F.L. performed laboratory analyses and provided analytical tools. S.F.B. analyzed the data. S.F.B. and M.G.A.v.d.H. wrote the first paper draft. S.F.B., D.B., L.B., J.H., E.K., N.P., R.D., S.M., D.M., P.V., A.B., F.L., S.G., Y.H., P.P., P.G., M.G., D.J., T.D.B., M.G.A.v.d.H. commented on the manuscript and provided feedback.

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ACKNOWLEDGEMENTS

The authors thank all participants, citizens, farmers, gardeners, and scientists who actively contributed to this project, especially P. Agassis, C. Albertella, F. Allaz, T. Amez-Droz, M. Angst, S. Banholzer, C. Basile, S. Baumann, Y. Berger, J. Bircher, U. Blum, M. Borghetto, A. BrownUlli, J. Brülhart, D. Buchmann, I. Bühlmann-Peterhans, M. Burgherr, A. Bürki, V. Buser, M. Cal, S. Christin, M. Cordier, B. Cuche, J.L. Dalle, A. Darbellay, D. Debrunner, P. Delco, A. Délétroz, D. Demiéville, M. Devay-Stäubli, A. Devittori, C. Diem, M. Dorsaz, S. Duc Emery, U. Erne, D. Fahrni, J.B. Fellay, F.G. Ferrari, G.R. Fischli, E. Fragnière, A. Frei, A. Gallmeister, C.N. Gerbex, E. Gianella, A. Gianinazzi, G. Giudici, N. Gloor, I. Graber, M. Gubler, L. Guerra, C. Guhl, G. Guignard, D. Guinans, H. Gysel, S. Hagenbuch, P. Haldimann, M. Hänseler, J. Häring, Z. Haubenschmid, T. Haussener, R. Hediger-Kämpf, C. Hofer, C.M. Hunger-Toggweiler, D. Hürlimann, B.L. Huser, W. Hutmacher, J.D. Knüsel, G. Köppel, P. Koster, M. Kuhn, C. Litscher, P. Lombriser, L. Maghetti, H. Mannale, A. Mapelli, L. Marchand, E. Marra, T. Neubauer, M. Nufer-Wolf, S. Pittaluga, C. Pittet, S. Rebetez, H. Reinhard, D. Reymond, H. Roder, P. Rohrer, F. Röösli-Probst, R. Rosa, P. Rüegg, R. Schamal, S. Schnyder, R. Schoepfer, S. Schürch, A. Schütz, R. Simian, L. Spycher, P. Streit, M. Sutter, H.A. Ulli-Brown, D. Valsesia, F. Veuthey, R. Vogel, J. Vogler, T. Vuilliomenet, R. Walder, M. Weingartner-Ritter, K. Wieser, L. Wirthner, N. Zaugg, A. Zaugg, and P. Zoppi. We are grateful for the wealth of interactions, questions, and feedback received. Moreover, the authors greatly thank the participants who participated in lab visits and sample preparation, including Cinzia Frei and her school class. We thank the Agroscope and University of Zurich Communication teams, the Agroscope translation service, the University of Zürich Postal office for their support, and all media, institutions, and people who helped spread the word for their interest. We thank Francis Martin for valuable input in planning the study. Moreover, we thank the Mercator foundation, the Ernst Göhner Foundation, the Swiss National Science Foundation (grant no. 194126), the Paul Schiller Foundation, the Foundation Dreiklang, and the Participatory Science Academy for funding the project. Open access publishing facilitated by Agroscope, as part of the Wiley - Agroscope agreement via the Consortium Of Swiss Academic Libraries.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in Zenodo (https://doi.org/10.5281/zenodo.20510023).

ETHICS STATEMENT

Participants voluntarily registered for the project and provided prior informed consent. They were free to withdraw from the study at any time without consequence. Personal data were collected solely for project administration and communication, stored securely, and handled in accordance with the EU General Data Protection Regulation (GDPR) and applicable national legislation. Participants could request the deletion of their data at any point. Any use of photographic material was subject to prior written consent from the individuals depicted. Only anonymized or pseudonymized data were used for scientific analyses and publications. The research focused exclusively on soil samples and associated environmental data; participants themselves were not the subjects of the research. Consequently, no human-subject research requiring ethics committee approval was conducted.

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How to cite this article: Bender, S. F., Bürge, D., Bragazza, L., Knop, E., Masson, S., Peter, N., Demarmels, R., Mueller, D., Imhof, A., Viviani, P., Bieri, A., Liebisch, F., Geisen, S., Hautier, Y., Panagos, P., Garcia-Palacios, P., Goberna, M., Johnson, D., Jöhr, S., … van der Heijden, M. G. A. (2026). Soil health assessment using buried cotton underpants with the help of 1000 citizen scientists. *Plants, People, Planet*, 1–19. https://doi.org/10.1002/ppp3.70259